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Confined Ionic Liquid in an Ionic Porous Aromatic Framework for Gas Separation

Ziqi Tian,^{1, 2} Sheng Dai,^{3, 4} De-en Jiang^{2,*}

¹Ningbo Institute of Materials Technology & Engineering, Chinese Academy of Sciences

1219 Zhongguan West Road, Zhenhai District, Ningbo 315201, PR China

²Department of Chemistry, University of California, Riverside, California 92521, United States

³Department of Chemistry, University of Tennessee, Knoxville, TN 37996

⁴Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831

*Corresponding author: djiang@ucr.edu

Abstract

Ionic liquids and porous frameworks are commonly employed in gas separation, but rarely together. To leverage their combined advantages, we design a novel system comprising an ionic liquid inside a charged porous-polymer framework. A positively charged ionic porous aromatic framework is impregnated with an ionic liquid, to confine the cations and anions inside the polymeric framework via electrostatic interactions. Molecular dynamics and free energy simulations are performed to determine the solubility, diffusivity, and hence permeability of CO₂ and CH₄ through the composite material. We find that the polymeric framework introduces nanoscale space in the confined IL, improving gas solubility, while the IL promotes gas transport in the porous structure. The confinement impacts CO₂ and CH₄ very differently regarding their solubility and diffusivity. By tuning the ionic-liquid loading, both high CO₂ permeability and high CO₂/CH₄ selectivity can be achieved, overcoming the Robeson upper bound. This work highlights the advantages of confined ionic liquids in an ionic porous-polymer framework in enhancing selective gas separation.

Keywords: ionic liquids, porous polymers, molecular dynamics, gas separations, membranes, confinement

Introduction

Efficient gas separations are essential to chemical industry and environmental remediation. Membrane-based technology has become particularly appealing due to their energy efficiency and small footprint.¹⁻³ Traditional membrane separation via polymers is governed by the solution-diffusion mechanism, leading to a tradeoff between permeability and selectivity called the Robeson upper bound.⁴⁻⁵ Materials with specific affinity to a permeate molecule and uniform nanoscale pores are supposed to circumvent the limit.⁶⁻⁷

Ionic liquids (ILs) are widely used in a supported liquid membrane for gas separation.⁸⁻⁹ They boast of negligible vapor pressure, good chemical and thermal stabilities, and great chemical tunability for gas separations.¹⁰⁻¹¹ But their liquid nature limits the membrane applications to low pressures. In addition, many ILs have high viscosity.

Porous materials such as metal organic framework (MOF)¹²⁻¹⁵ and covalent organic framework (COF)¹⁶⁻¹⁷ are popular sorbents for gas separations. Porous aromatic framework (PAF) as a porous organic polymer also attracted great interest.¹⁸ These materials possess large free volume and relatively uniform pore size ranging from several angstroms to few nanometers. Recently, much effort has been devoted to developing membranes for these porous materials. For instance, incorporating MOF-based nanostructures into polymeric matrices endowed composite membranes with outstanding gas separation performance.¹⁹⁻²¹ COF and MOF membranes could also grow on a porous substrate via surface fabrication.²²⁻²³ By controlling the thickness and component of the coating layer, gas separation would be significantly improved. Further, nanosheets with suitable aperture were exfoliated from bulk layered MOFs that could separate gas molecules by molecular sieving.²⁴⁻²⁵

Combining ILs and porous materials can lead to new avenues for discovering separation media. For example, a recent computational proof-of-concept demonstrated ion-gated gas separation through an IL/porous-graphene composite membrane where the ions of the IL are confined on a two-dimensional surface.²⁶ Three-dimensional porous frameworks can offer even more flexibility in confining the ions and tune the pore size and the interaction for gas selectivity.²⁷ Some MOF-IL composites have been designed theoretically²⁸⁻³² and synthesized experimentally³³⁻³⁷ for membrane gas separation. For example, Yang et al. prepared IL/ZIF-8 composites that have high CO₂/CH₄ selectivity and CO₂ permeability.³³ A more recent study showed that the IL/ZIF-8 core-shell composite can dramatically increase CO₂/CH₄ adsorption selectivity.³⁸

A critical concern in preparing the composite of a fluid and a nanoporous solid is how to disperse the liquid evenly in the pores of the framework. To this end, interaction between the IL and the porous framework has to be strengthened as much as possible. Given the ionic nature of the IL, one obvious idea would be to use an ionic porous framework to maximize the electrostatic interaction with the confined IL. Recently a series of ionic covalent frameworks have been reported,³⁹⁻⁴³ which provide an ideal micro-environment to confine an IL for gas separations. A typical such ionic covalent framework is quaternary-phosphonium-functionalized ionic porous aromatic framework (iPAF). Hence an IL confined in an ionic porous polymer such as iPAF would provide an ideal system to understand how the confined ions in a 3D framework modulate gas separation. To this end, herein we use the iPAF as the porous framework to build up the IL-iPAF composite systems and simulate their performance for CO₂/CH₄ membrane separation, in an effort to shed light on how the gas solubility and diffusivity are affected and can be tuned in the composite systems.

Simulation models and methods

By using sp³-hybridized phosphorous atoms to replace the sp³-C nodes in the ideal diamond-like PAF-1 structure (Figure 1a),⁴⁴ the iPAF structure is obtained (Figure 1b). The positive charges on the P atoms are balanced by the counter anions (green balls in Figure 1b). Structure and cell size of the iPAF with the TFSI anion were optimized with density functional theory (DFT) using Vienna *ab initio* simulation package (VASP).⁴⁵ The Perdew-Burke-Erzerhof (PBE) functional⁴⁶ was employed with projector-augmented wave method (PAW)⁴⁷ to describe the electronic structure. Energy cutoff was set to be 400 eV and only the Γ -point was used to sample the Brillouin zone. Since in the iPAF and IL-iPAF systems the electrostatic interaction dominates over the van der Waals interaction, we did not include dispersion correction in our DFT geometry optimizations.

A commonly studied IL, [bmim][TFSI] (Figure 1c), was chosen to fill into the porous structure of the iPAF-TFSI system. We examined various IL uptakes, from 0 to 18 ion pairs per unit cell. Molecular dynamics (MD) simulations were carried out on the 2×2×2 supercell model with Lammmps under the NPT ensemble.⁴⁸ We used the FFtool code to assist in generating parameter files of the OPLS-AA type.⁴⁹ All the force field parameters are provided in the Supporting Information. Atomic charges were scaled by 0.8 to consider polarization and charge transfer, also

in accordance with fitted charges of framework. EPM2 models were used to simulate gases.⁵⁰ The electrostatic interaction was summed with the PPPM method. The cutoff for non-bonded interactions was 12 Å and long-range corrections were applied. Simulated systems were heated to 800 K for 10 ns and quenched to 300 K in 10 ns. Subsequently equilibrium MD simulations were run at 300 K for 25 ns. Nose-Hoover thermostat was employed to control temperature with a damping constant of 100 ps while the Nose-Hoover barostat was used to maintain pressure at 1 atm with a damping constant of 500 ps. For comparison, the bulk IL was simulated with the same force-field and control parameters for a cubic-box supercell containing 100 ion pairs.

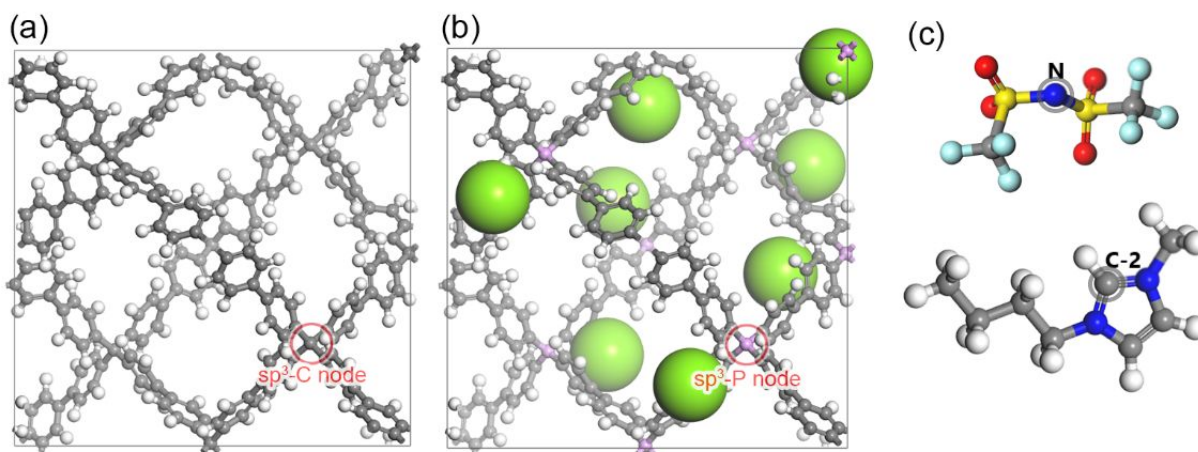


Figure 1. (a) Structure of the ideal diamond-like PAF-1: the bond between two sp^3 -C atoms in the diamond structure is replaced by the biphenyl unit. (b) The structure of iPAF studied in this work: P-atom nodes (pink) replace the sp^3 -C nodes in PAF-1; each positively charged P node is balanced by an anion (large green ball). (c) Structures of [TFSI] anion and [bmim] cation: the N atom in the anion and the C-2 atom in the cation are marked for radial distribution function analysis later. Color code: white, H; grey, C; blue, N; red, O; cyan, F; yellow, S; pink, P.

Gas diffusivity (D) in the composite system was obtained from the slope of the mean squared displacement (MSD) in the MD trajectory. To avoid random error, twenty paralleled 25 ns-MD simulations were performed for each system. Previous simulations indicated that the gas loading had little influence on MSD,⁵¹⁻⁵² so we placed 10 gas molecules in each simulation box. Next, free energy perturbation (FEP) approach was implemented to compute gas solvation free energy, ΔG_{sol} , and then to estimate solubility (S):

$$S = \frac{1}{RT} \exp \left(- \frac{\Delta G_{sol}}{RT} \right) \quad (1).$$

The soft-core potential for gas was used to avoid singularity issues. Coupling parameter λ varied from both 0 to 1 and 1 to 0 at an interval of 0.0025, so 400 intermediate states in total; for each λ value or state, 100 ps of simulation was run for a total simulation time of 40 ns. Multiple independent runs were carried out to obtain average ΔG_{sol} .

To validate parameters and the method, we first computed the permeability ($P = D \times S$) and selectivity in bulk [bmim][TFSI]. Some important properties, such as the Henry’s Law constant (H) and gas permeability, have been measured experimentally.^{33, 55} We compared the simulated values with available experimental data in **Table 1**. One can see that our simulations reproduce permeability and selectivity in good agreement with experiment. We further applied this approach to predict the gas separation performance in [bmim][TFSI]/ZIF-8 and found that the predicted CO₂/CH₄ selectivity (40.5) is in good agreement with the experiment (30 ~ 40).³³

Table 1. Comparison of simulated and experimental Henry’s Law Constant (H), diffusivity (D), permeability (P), and selectivity (S) of CO₂ and CH₄ in [bmim][TFSI].

	H _{CO2} (MPa)	D _{CO2} (Å ² /ns)	H _{CH4} (MPa)	D _{CH4} (Å ² /ns)	P _{CO2} (barrer)	P _{CH4} (barrer)	S _{CO2/CH4}
Exp. ^a	3.5	58	52.4	-	1344	110	12.2
Sim.	3.40	27.8	66.5	22.6	833	34.6	24.1

^a Ref. 55.

Results and discussion

Structure of iPAF-TFSI. We started with the diamond-like structure for iPAF-TFSI (Figure 2a) with a lattice parameter of 24.2 Å based on the parent PAF-1 structure (Figure 1a). Then we allowed the lattice parameters to change and obtained a new distorted iPAF-TFSI structure with a much smaller lattice parameter of 17.8 Å (Figure 2b). Both the force-field-based and DFT-based calculations found the distorted structure energetically more favorable than the diamond-like structure, by 4.40 and 2.36 eV/cell, respectively. The optimized, distorted structure has a much smaller free volume. Previous experimental studies also indicated more condensed networks for

some PAF derivatives than the ideal diamond topology.⁵⁶ The iPAF-IL composites examined below will be therefore based on the more stable, distorted structure of iPAF-TFSI and the popular IL, [bmim][TFSI] (Figure 1c).

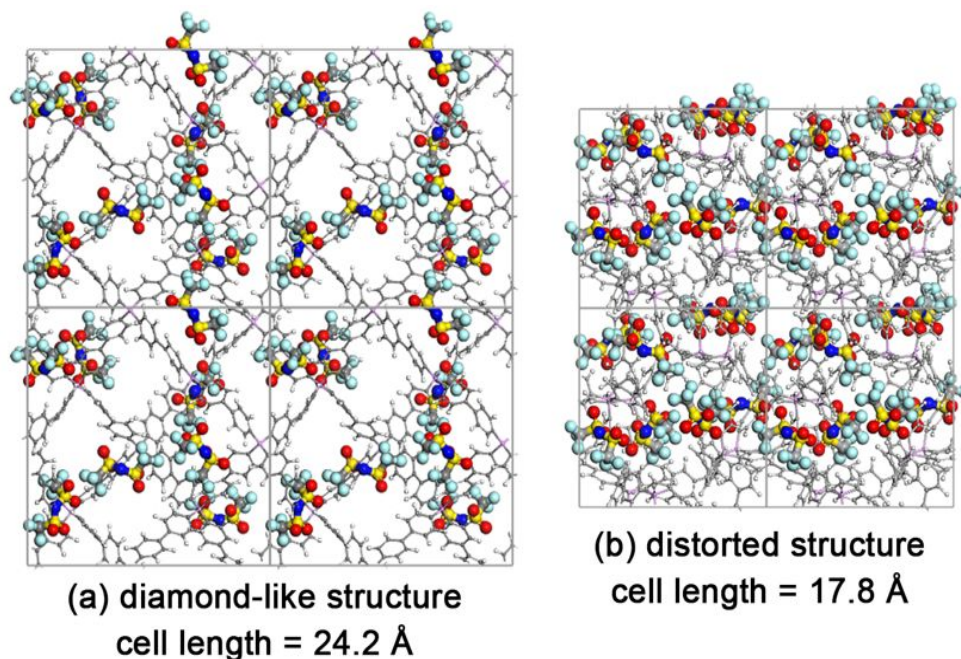


Figure 2. (a) The initial diamond-like iPAF-TFSI structure; (b) the distorted and shrunken structure after DFT geometry optimization. The structures are shown in a front view of a $2 \times 2 \times 2$ supercell; the iPAF framework in the line model, the TFSI anions in the CPK model. A Color code is the same as in Figure 1.

Structure and properties of the iPAF-[bmim][TFSI] composite. We next simulated the composite system with various uptakes of the [bmim][TFSI] IL. Because the parent iPAF has a cubic initial structure and we used a large cell of many ion pairs inside the iPAF, we assume that the IL-iPAF systems will be in general isotropic. So in our NPT simulations, we maintained the cubic cell shape but allowed only the cell size or volume to vary. We think that this is a good approximation to generate reproducible and consistent gas solubility and diffusivity data inside the IL-iPAF systems. For simplicity, the composite containing n IL pairs per unit cell is labeled as iPAF- n IL. Figure 3 shows the structures of the iPAF- n IL systems for $n = 4, 8, 12$, and 16 , corresponding to P/[bmim] ratios of 0.5, 1.0, 1.5 and 2.0, respectively. One can see the gradually

increasing lattice parameter, as more IL ion pairs are loaded into the iPAF, similar to the swelling behavior of a polymer. We found that as many as 16 IL pairs can be fully loaded into the iPAF framework. For modeling of gas permeation, we considered three composite systems with relatively high IL loadings, i. e., iPAF-8IL, iPAF-12IL, and iPAF-16IL.

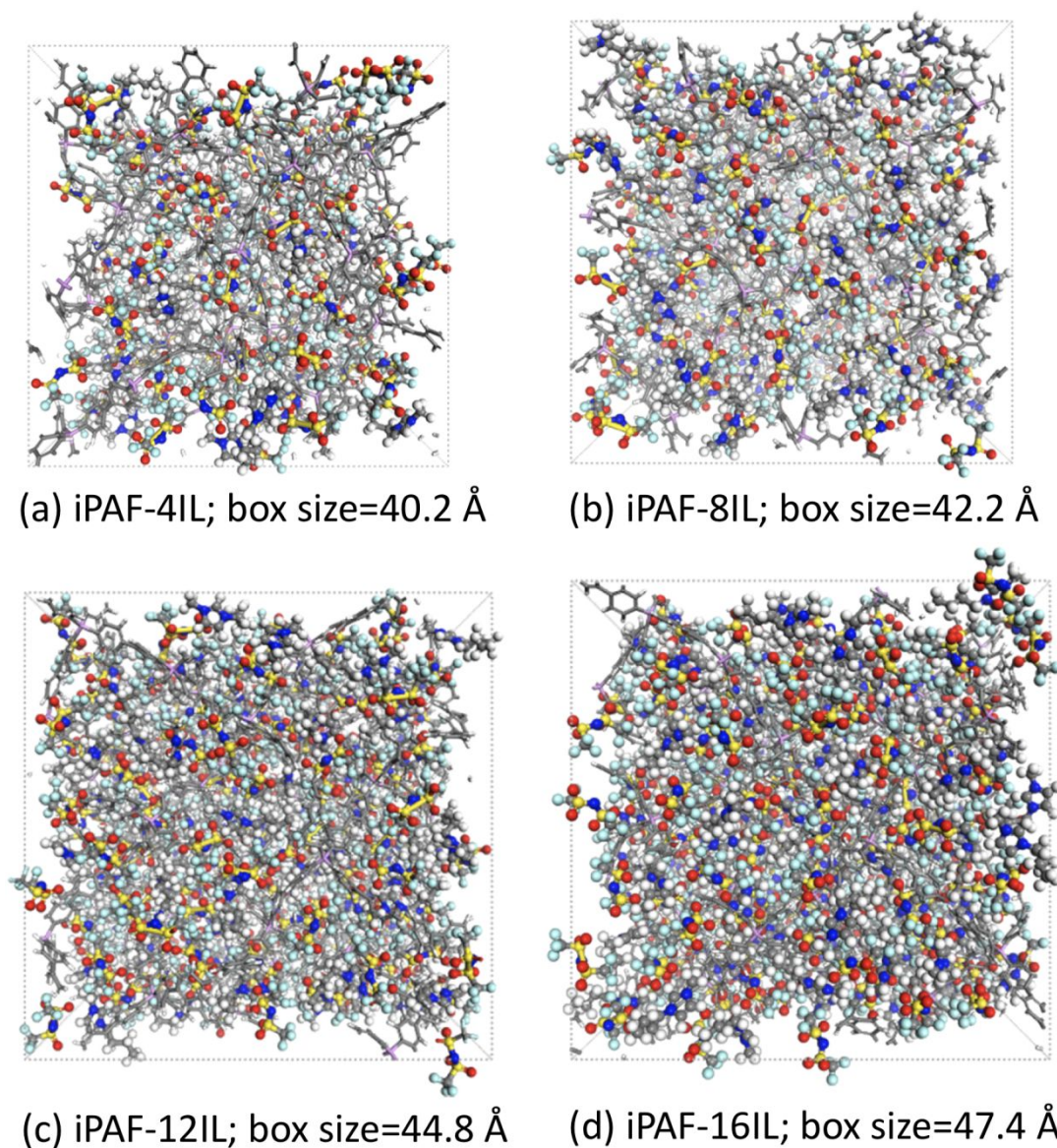


Figure 3. The snapshots of the iPAF-*n*IL composite systems: (a) *n* = 4; (b) *n* = 8; (c) *n* = 12; (d) *n* = 16. The corresponding IL/iPAF weight ratios are 0.625, 1.25, 1.875 and 2.50, respectively; the corresponding P/[bmim] ratios are 0.5, 1.0, 1.5 and 2.0, respectively.

Radial distributions functions (RDFs) of ions to the phosphorous node in the composites are plotted in Figure 4. The smooth and broad RDF curves of ions indicate the fluid nature of the IL in the confined environment. Because the positive charges of the iPAF framework are mainly centered on the P atoms, so they attract the [TFSI]⁻ anions and repel the [bmim]⁺ cations, and the RDFs in Figure 4 show that the [TFSI]⁻ anions (dashed lines) are closer than the [bmim]⁺ cations (solid lines) are to the P atoms in iPAF. Compared to the peaks of the [TFSI]⁻ anions (~ 7.0 Å), the distributions of the [bmim]⁺ cations are broader (8 to 10 Å) and shift to the right with the IL loading, indicating a weakened interaction between the cations and the iPAF with the IL loading.

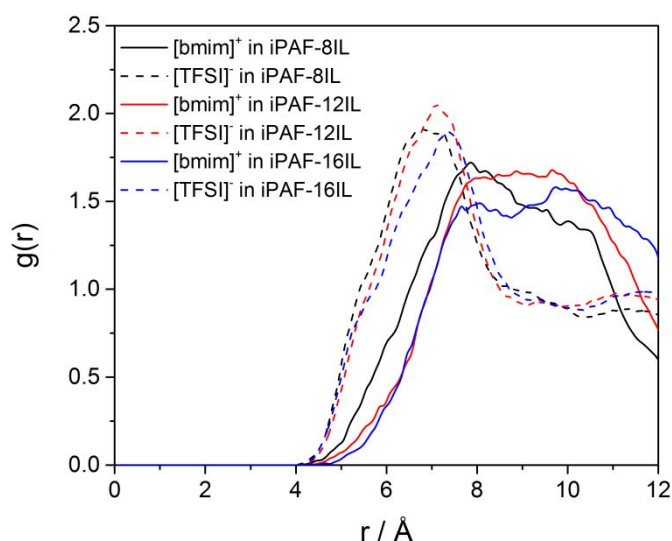


Figure 4. Radial distributions functions (RDF) of [bmim]⁺ cations (from the C-2 position on the imidazolium ring; Figure 1c) and [TFSI]⁻ anions (from the N atom; Figure 1c) around the phosphorous node in the positively charged iPAF framework for iPAF-IL composites of different IL loadings.

Mean squared displacements (MSDs) of ions are plotted in Figure 5. One can see that the [TFSI]⁻ anions are rather immobile in the iPAF-TFSI framework itself, due to the strong attraction by the positively charged framework. As shown in Figure S1, the MSDs of cation and anion in iPAF-4IL are quite similar to the MSD of anion in iPAF-TFSI; in other words, the ions are tightly trapped in the pores when the loading is low. On the other hand, both cations and anions are mobile in the iPAF-IL composites with higher IL loadings at the simulation temperature (300 K), even

though they move much slower than ions in the bulk ionic liquid. Because anions are more strongly attracted to the positively charged framework, the $[\text{TFSI}]^-$ anions move much slower than the $[\text{bmim}]^+$ cations in the same composite. Interestingly, the anion MSDs are almost identical in the three composites of different IL loadings, but the MSD of the cation increases with the loading of the ionic liquid. From iPAF-8IL to iPAF-16IL, the cation diffusivity increases from $0.0787 \text{ \AA}^2/\text{ns}$ to $0.121 \text{ \AA}^2/\text{ns}$. The increasing cation diffusivity correlates with both the lattice expansion (Figure 3) and the weakened cation-framework interaction (Figure 4) with the IL loading. The higher mobility of the cations can help the transport of the gas molecules through the iPAF-IL composite.

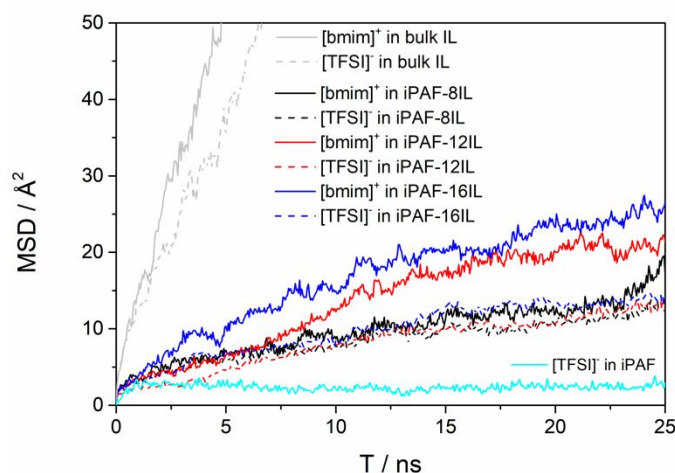


Figure 5. Mean squared displacements (MSDs) of cations and anions in the iPAF-[bmim][TFSI] composites of various IL loadings. For comparison, MSDs in the bulk IL and iPAF are also shown.

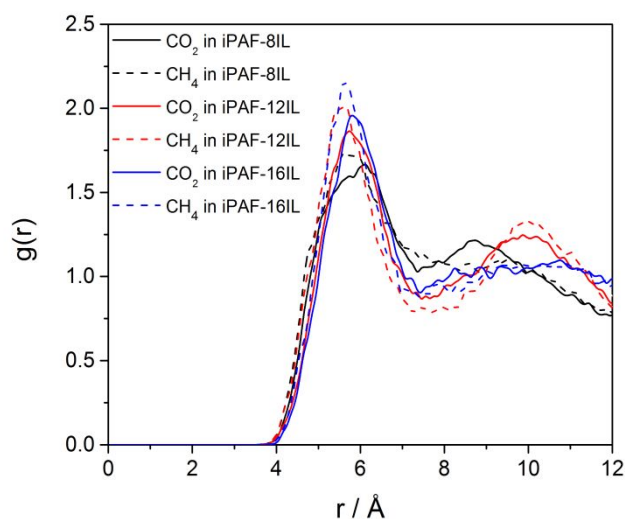


Figure 6. RDFs of gases (from the Carbon atom) to the positively charged P nodes of iPAF.

Structure of CO₂ and CH₄ in the iPAF-[bmim][TFSI] composite. To examine the performance of the iPAF-IL composite for CO₂/CH₄ separation, we first investigated the structure of CO₂ and CH₄ in the composites of different IL loadings. Figure 6 shows that the RDFs of CO₂ and CH₄ around the phosphorous nodes of the iPAF are almost the same at various loadings: the average positions of the first peaks for CO₂ and CH₄ are at 5.78 Å and 5.62 Å, respectively. Figure 7 plots the RDFs of gases around cations and anions. CO₂ has some closer contact with the anions than with the cations, while CH₄ has more probability around the anions than around the cations. Typical snapshots of gas molecules in the iPAF-12IL composite system are shown in Figure 8.

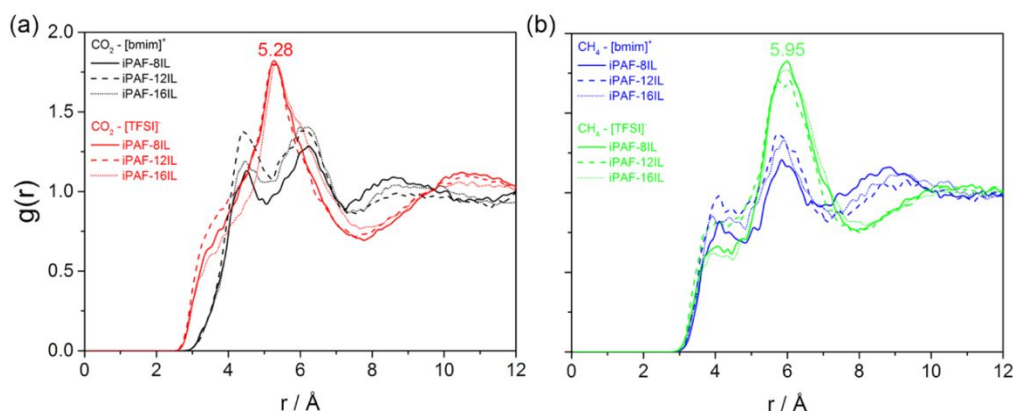


Figure 7. RDFs of (a) CO₂ and (b) CH₄ to the cations (represented by the C-2 position on the imidazolium ring) and anions (represented by the N atom) in composites with various IL loadings.

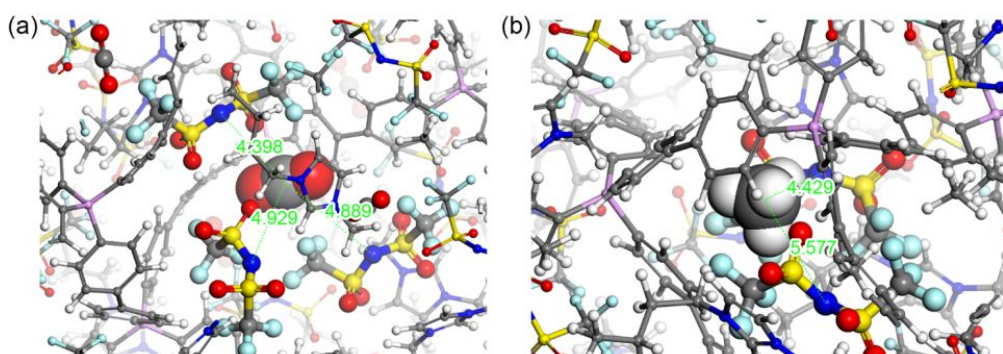


Figure 8. Typical snapshots of (a) CO₂ and (b) CH₄ molecules inside the iPAF-12IL composite.

Performance of the iPAF-[bmim][TFSI] composite for CO₂/CH₄ separation. To estimate the gas permeability of the iPAF-[bmim][TFSI] composite membrane, we first simulated the solvation free energy, ΔG_{sol} , and solubility (S) of CO₂ and CH₄ separation in the iPAF-[bmim][TFSI] composites of different IL loadings. Table 2 lists all simulated ΔG_{sol} of CO₂ and CH₄ in the iPAF-[bmim][TFSI] composites of various IL loadings. We then converted ΔG_{sol} to gas solubility at 298 K and 1 bar (Table 3). One can see that both CO₂ and CH₄ solubilities are higher in the composite than in the bulk IL, while CH₄ solubility is much lower in the composite than in the iPAF itself, leading to a much higher solubility selectivity (SS) of CO₂/CH₄ in the composite than in the iPAF (Table 3). Interestingly, the iPAF-8IL composite's CO₂/CH₄ solubility selectivity is even higher than that of the bulk IL.

Table 2. Simulated solvation free energy (ΔG_{sol}) of CO₂ and CH₄ in the iPAF-[bmim][TFSI] composites of various IL loadings. For comparison, values in the bulk IL and the iPAF itself are also shown.

	$\Delta G_{\text{sol}}(\text{CO}_2)$ (kcal/mol)	$\Delta G_{\text{sol}}(\text{CH}_4)$ (kcal/mol)
Bulk IL	-0.55 ± 0.08	1.22 ± 0.05
iPAF-16IL	-1.29 ± 0.19	0.36 ± 0.04
iPAF-12IL	-1.35 ± 0.04	0.47 ± 0.16
iPAF-8IL	-1.45 ± 0.23	0.89 ± 0.04
iPAF	-1.70 ± 0.15	-0.91 ± 0.18

Table 3. Simulated solubility (S), diffusivity (D), and permeability (P) of CO₂ and CH₄ as well as their CO₂/CH₄ selectivities (SS, DS, and PS) in the iPAF-[bmim][TFSI] composites of various IL loadings. For comparison, values in the bulk IL and the iPAF itself are also shown. Uncertainties ($\pm$) are given in parentheses.

	S_{CO_2} (mol/L ⁻¹ bar ⁻¹)	S_{CH_4} (mol/L ⁻¹ bar ⁻¹)	SS	D_{CO_2} (Å ² /ns)	D_{CH_4} (Å ² /ns)	DS	P_{CO_2} (Barrer)	P_{CH_4} (Barrer)	PS
Bulk IL	0.102 (0.013)	0.0041 (0.0004)	24.9 (4.2)	27.8 (0.10)	22.6 (0.08)	1.23 (0.01)	834 (112)	34.6 (2.9)	24.1 (3.8)
iPAF-16IL	0.351 (0.112)	0.022 (0.002)	16.0 (6.3)	14.3 (0.09)	7.83 (0.04)	1.83 (0.02)	1479 (482)	50.9 (3.6)	29.1 (9.7)
iPAF-12IL	0.393 (0.025)	0.018 (0.005)	21.8 (6.9)	17.4 (0.05)	9.49 (0.04)	1.83 (0.01)	2018 (132)	51.7 (13.6)	39.0 (10.6)

iPAF-8IL	0.466	0.009	51.8	7.30	1.67	4.37	1002	4.5	223
	(0.174)	(0.001)	(22.3)	(0.04)	(0.08)	(0.23)	(384)	(0.35)	(86.9)
iPAF	0.704	0.187	3.76	1.71	8.01	0.21	352	439	0.80
	(0.173)	(0.055)	(2.07)	(0.02)	(0.03)	(0.003)	(96)	(89)	(0.31)

Table 3 further compares diffusivity (D) and permeability ($P = D \times S$) of CO_2 and CH_4 in the iPAF-[bmim][TFSI] composites as well as the selectivity. In the iPAF itself, CO_2 is adsorbed strongly inside the sorbent, leading to a low gas diffusivity. With the IL loading, the gas diffusivity increases and there exists an optimal loading of iPAF-12IL. Although gas diffusivity is still lower than in the bulk IL, the CO_2/CH_4 diffusivity selectivity (DS) is actually higher in the iPAF-IL composite than in the bulk IL, because the confined ions slow down CH_4 more than they do CO_2 . The combined effect of solubility and diffusivity is a higher permeability of CO_2 and a higher CO_2/CH_4 selectivity in the composite than in either the bulk IL or the iPAF itself. We plot CO_2 permeability versus CO_2/CH_4 selectivity in Figure 9, together with the 2008 Robeson upper bound.⁴ The simulated data for the bulk IL are in good agreement with the experiment. Compared with the bulk IL and the iPAF, all three composite systems show improved separation performances above the upper bound from our simulations. Especially, the iPAF-8IL system is most promising, because at this loading both the solubility selectivity and the diffusivity selectivity are the highest (Table 3).

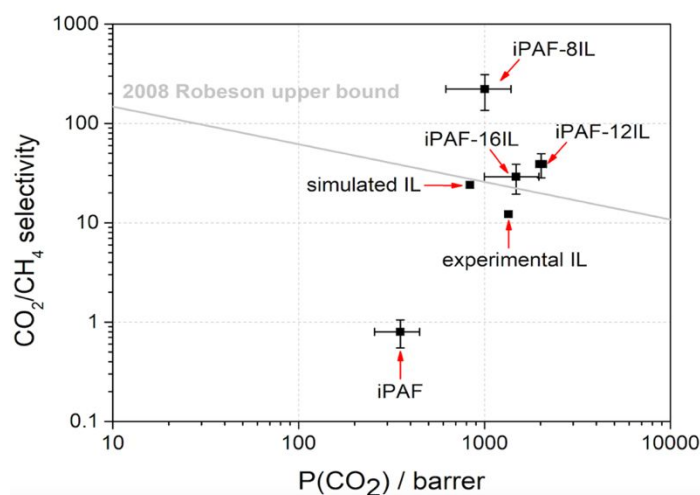


Figure 9. CO_2 permeability versus CO_2/CH_4 selectivity of composite with various ionic liquid loading.

Implications of our simulation results. Our simulation results in Table 3 and Figure 9 suggest a promising way to overcome the Robeson upper bound by using the iPAF/IL composite. Loading can be used as a parameter to further tune the performance. Recent experiments have demonstrated the advantages in combining MOFs and ILs for gas separation.^{33, 38, 57} Here we further show that a charged porous organic framework loaded with ILs holds the potential for breakthrough membrane performances. We think that this idea can also be applied to the porous organic polymers that do not necessarily have a well-defined periodic structure.

Conclusion

In summary, we have designed and simulated a composite material made of an ionic liquid confined inside an ionic porous-polymer aromatic framework (iPAF-IL) for membrane gas separation of CO₂ and CH₄. We optimized the structure with density functional theory and performed classical molecular dynamics and free-energy simulations to determine the solubility, diffusivity, and hence permeability of CO₂ and CH₄ in the composite material of different IL loadings. We found that gas solubility is higher in the composite than in the bulk IL and it is more so for CO₂ than for CH₄, while gas diffusivity is lower in the composite than in the bulk IL and the confined ions slow down CO₂ less than they do CH₄. As a result, both higher CO₂ permeability and higher CO₂/CH₄ selectivity could be achieved in the composite than in the bulk IL or the iPAF itself. For certain loading, the simulated performance is significantly above the Robeson upper bound. This work shows the promise of confined ionic liquids in porous organic polymers in improving membrane gas separation.

Supporting Information

Additional figures; force field parameter; coordinates.

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